CURRICULUM VITAE

Jaime Castresana Iza

Software Engineer | Product Engineering, Applied AI & Audio



Software engineer building full-stack engineering software and independent AI applications. I combine user discovery, backend and frontend implementation, testing, and an audio/DSP foundation to turn complex data and specialist workflows into useful, reliable products.

CONTACT

EMAIL
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LOCATION
Copenhagen, Denmark

GITHUB
github.com/jci24

WEBSITE
jaimecastresana.com

CORE COMPETENCIES

PRODUCT ENGINEERING

Technical product development · Workflow mapping and UX · Testing and validation · POC-to-release delivery

AUDIO & SIGNAL

Acoustic analysis · Sound quality · Embedded audio and DSP · Measurement workflows

AI & AUTOMATION

LLM application development · Tool calling · Agent-style workflows · Evidence-bound evaluation

CORE TOOLS

C#/.NET · React · TypeScript · Python · MATLAB · C/C++ · Git · CI/CD · REST APIs · Azure DevOps

DEVELOPMENT

Selected by management for the one-week Effective Communication Skills for Professionals programme.

BEYOND WORK

Music producer for independent artists in Spain, including Suave, winner of VillaSound Festival Bilbao 2021.

LANGUAGES

Spanish - Native
English - C1
Danish - A2
French - B1

PROFESSIONAL EXPERIENCE

Hottinger Brüel & Kjær

Sep 2023 - Present

SOFTWARE ENGINEER

- Owned product features across requirements, technical design, C#/.NET APIs, React implementation, automated tests, and delivery.
- Gathered user and specialist requirements; separated essential from optional needs before implementation.
- Compared UI/UX, algorithm, and visualisation-library options; documented trade-offs and presented decisions to stakeholders.
- Ran user tests and specialist reviews for acoustic model-comparison, visualisation, and validation workflows.

ICEpower A/S

May 2022 - Jun 2023

R&D SOFTWARE ASSISTANT

- Extended an existing Python and pytest framework for amplifier, network, audio, and communication testing.
- Tested traffic stress, disruption, power cycling, triggering, and recovery; processed results and generated plots in Python.
- Contributed automation to existing test pipelines and investigated system behaviour across failure and recovery paths.
- Modelled digital crossover responses in MATLAB and implemented selected filters in C++ on ARM Cortex-M7.

EDUCATION

MSc Sound and Music Computing

2021 - 2023

Aalborg University, Copenhagen Campus

BSc Industrial Technology Engineering

2015 - 2020

University of Deusto

SELECTED PROJECT

JustCall

INDEPENDENT PROJECT

RELIABLE VOICE PRACTICE PRODUCT

jaimecastresana.com/projects/justcall

- React Native app with authenticated TypeScript API, voice integration, owned session evidence, and explicit recovery paths.
- Validated through automated tests and ten consecutive two-way iPhone calls; production and learner outcomes remain separate gates.

TypeScript · React Native · Hono · Supabase · Voice AI

FROM VOICE DEMO TO TESTED SESSION RELIABILITY